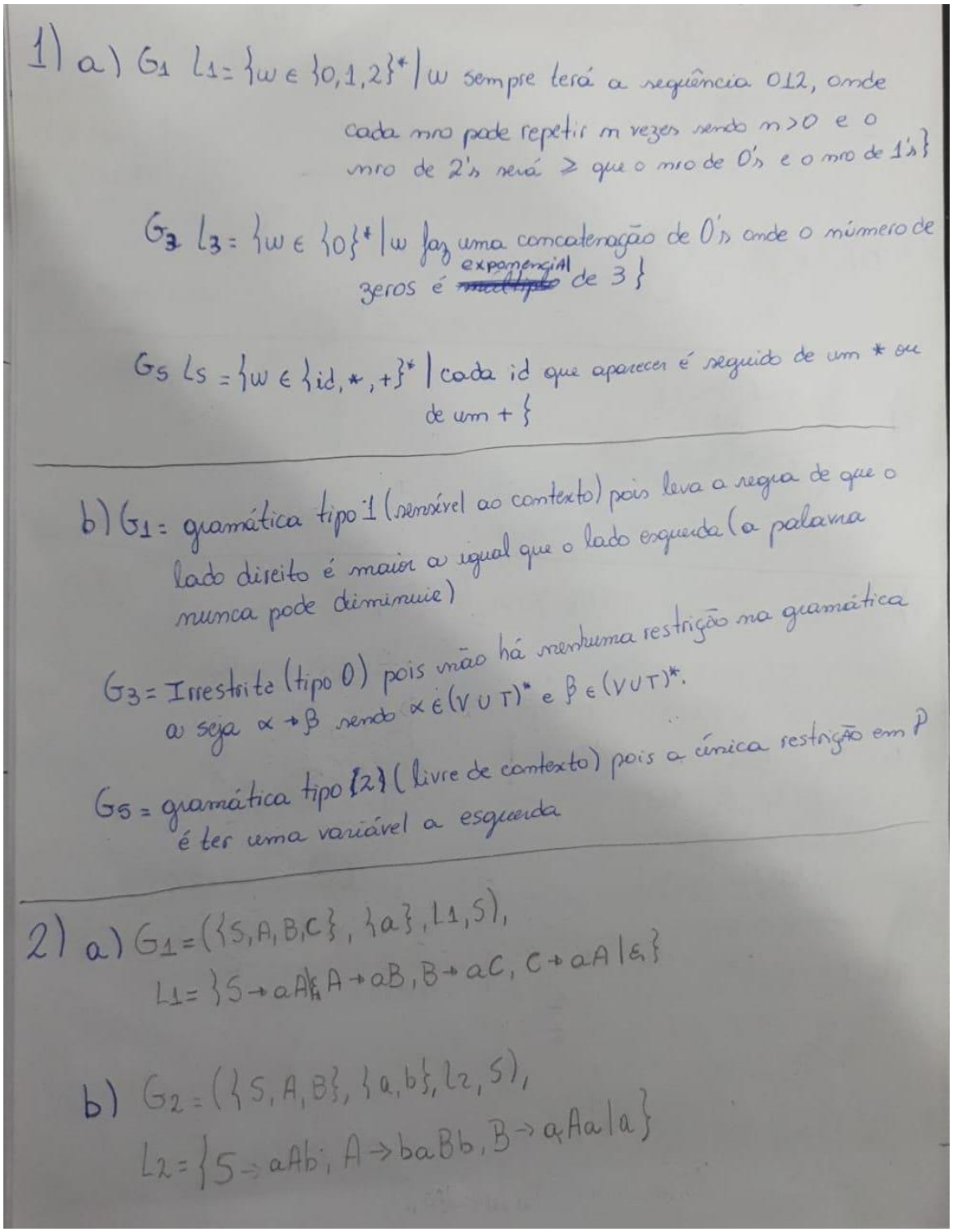
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Matrícula: 11721BCC029

LFA Lista 1



1) a) G_1 $L_1 = \{w \in \{0,1,2\}^* \mid w$ sempre terá a sequência 012, onde cada nro pode repetir n vezes sendo $n>0$ e o nro de 2's será $\geq$ que o nro de 0's e o nro de 1's$\}$

G_3 $L_3 = \{w \in \{0\}^* \mid w$ faz uma concatenação de 0's onde o número de zeros é ~~multiplo~~ exponencial de 3$\}$

G_5 $L_5 = \{w \in \{id, *, +\}^* \mid$ cada id que aparecer é seguido de um * ou de um +$\}$

b) G_1 = gramática tipo 1 (sensível ao contexto) pois leva a regra de que o lado direito é maior ou igual que o lado esquerda (a palavra nunca pode diminuir)

G_3 = Irrestrita (tipo 0) pois não há nenhuma restrição na gramática ou seja $\alpha \to \beta$ sendo $\alpha \in (V \cup T)^*$ e $\beta \in (V \cup T)^*$.

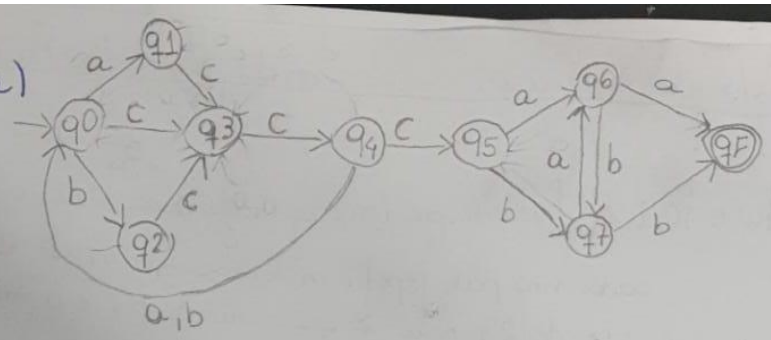
G_5 = gramática tipo 2 (livre de contexto) pois a única restrição em P é ter uma variável a esquerda

2) a) $G_1 = (\{S, A, B, C\}, \{a\}, L_1, S)$,
$L_1 = \{S \to aA, A \to aB, B \to aC, C \to aA \mid \varepsilon\}$

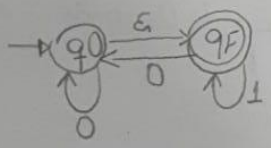
b) $G_2 = (\{S, A, B\}, \{a, b\}, L_2, S)$,
$L_2 = \{S \to aAb, A \to baBb, B \to aAa \mid a\}$

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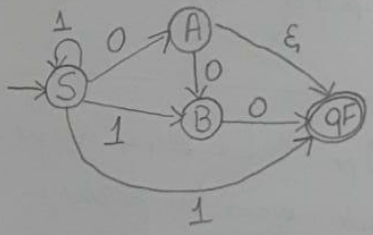
3) a)



b)



4)



5) 1° passo: converter o AFNε para AFND

$\delta'(q_0,0) = \underline{\delta}(\{q_0\},0) = \{q_3,q_1\}$

$\delta'(q_0,1) = \underline{\delta}(\{q_0\},1) = \{q_0,q_1,q_3\}$

$\delta'(q_1,0)$ e $\delta'(q_1,1)$ são indeterminados

$\delta'(q_2,0) = \underline{\delta}(\{q_2\},0) = \{q_1\}$

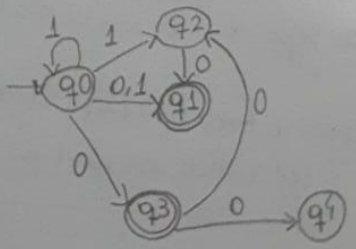
$\delta'(q_2,1)$ = indeterminado

$\delta'(q_3,0) = \underline{\delta}(\{q_3\},0) = \{q_4,q_2\}$

$\delta'(q_3,1)$ = indeterminado

$\delta'(q_4,0)$ e $\delta'(q_4,1)$ não indeterminados

AFND resultante:



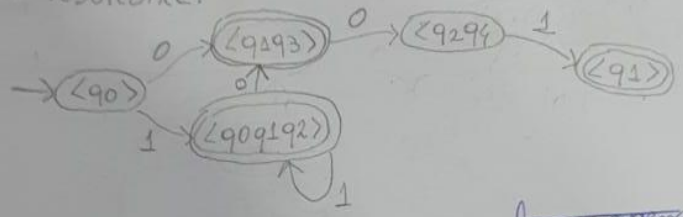
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2° Passo: converter o AFND para AFD

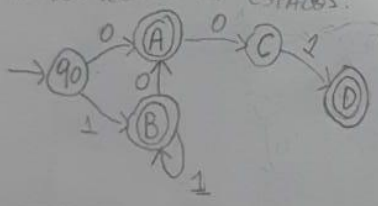
δ	0	1
→q0	{q1,q3}	{q0,q1,q2}
*q1	–	–
q2	{q1}	–
*q3	{q2,q4}	–
q4	–	–

δ'	0	1
<q0>	<q1q3>	<q0q1q2>
*<q1q3>	<q2q4>	<–>
*<q0q1q2>	<q1q3>	<q0q1q2>
<q2q4>	<q1>	<–>
*<q1>	<–>	<–>

AFD resultante:



Renomeando os estados:



~~linguagem gerada~~

gramática:

G1 = ({S, A, B, C, D}, {0,1}, L1, S)

L1 = {S → 0A | 1B, A → ε | 0C, B → ε | 0A | 1B, C → 1D, D → ε}

estado final (FLCA-N) (todos no norte)

6) a) (FLCA-S) = todos na margem sul

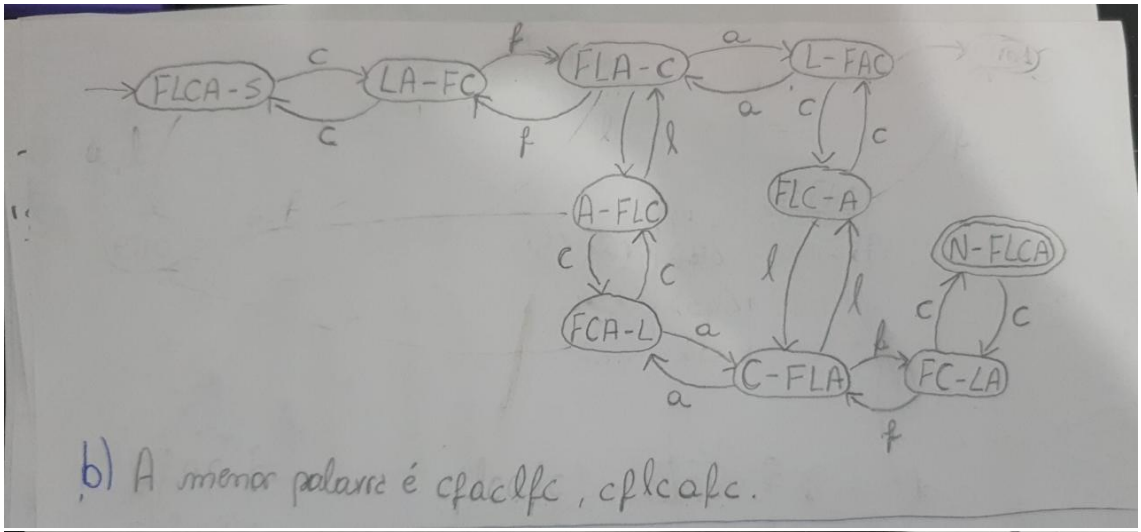
(L-FCA) = lobo no sul, fazendeiro, cabra, alface no norte

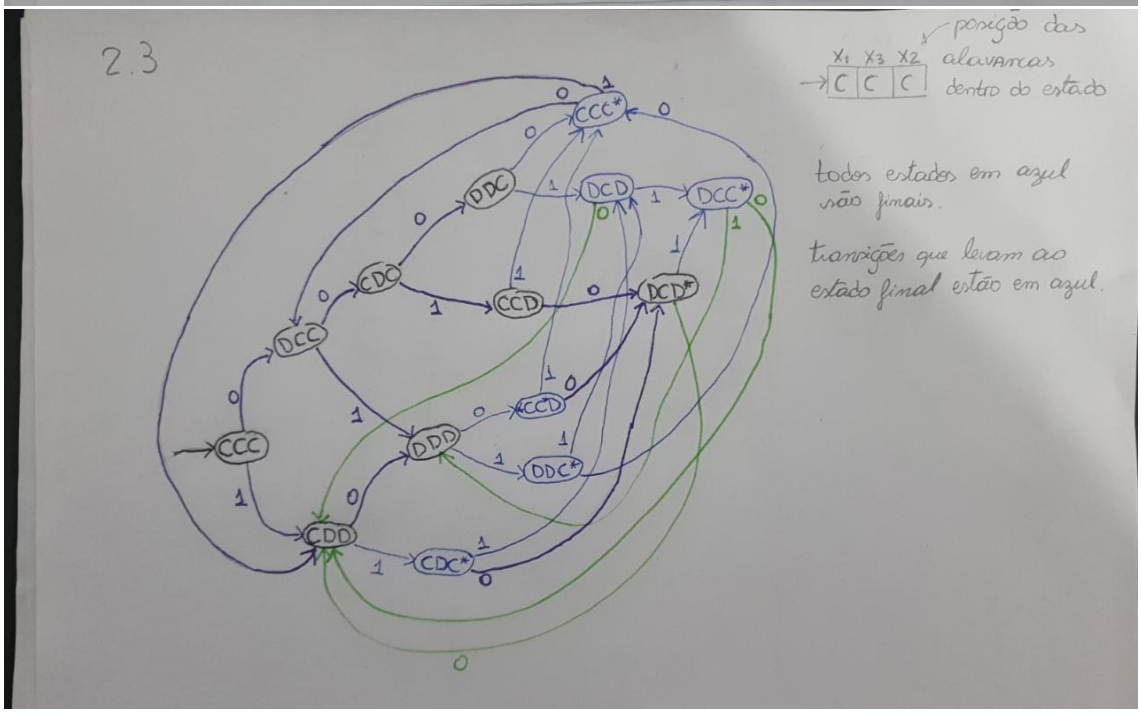
Entradas:

f - fazendeiro atravessa sozinho
l - fazendeiro atravessa rio c/ lobo
c - fazendeiro atravessa rio c/ cabra
a - fazendeiro atravessa rio c/ alface

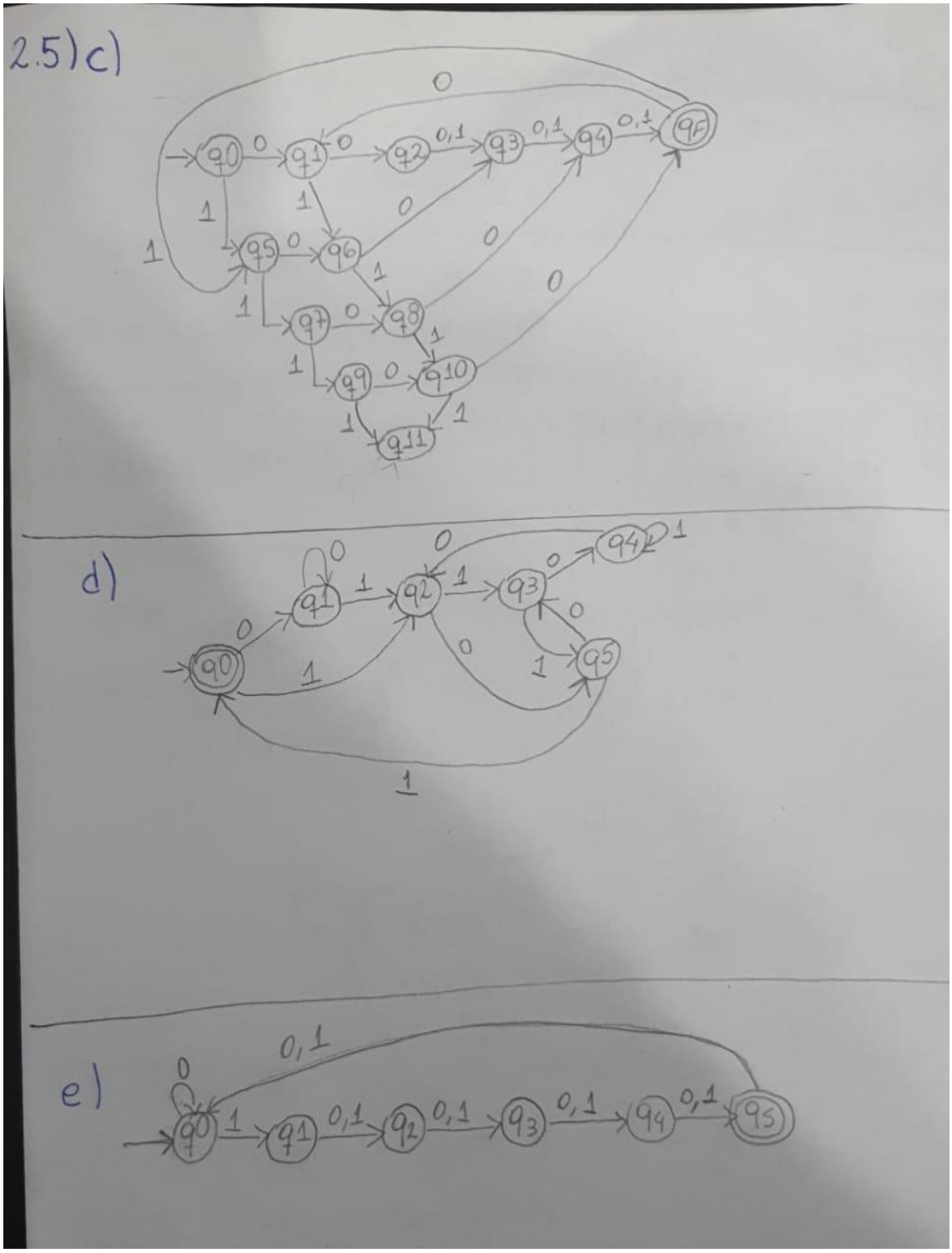
estados serão separados por 1 tiago
as letras representam o fazendeiro (F), lobo (L), cabra (C), alface (A).
do lado esquerdo do tiago representa quem ficou no sul, do lado direito quem ficou no norte.

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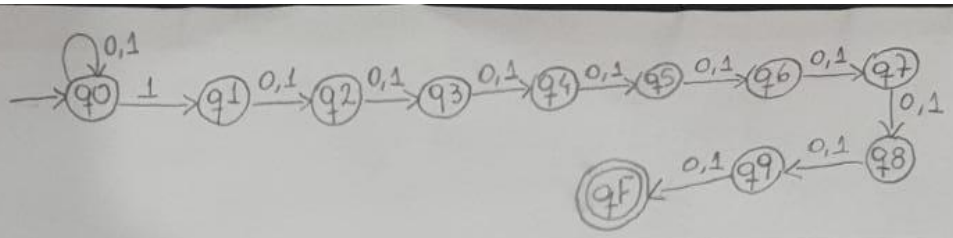


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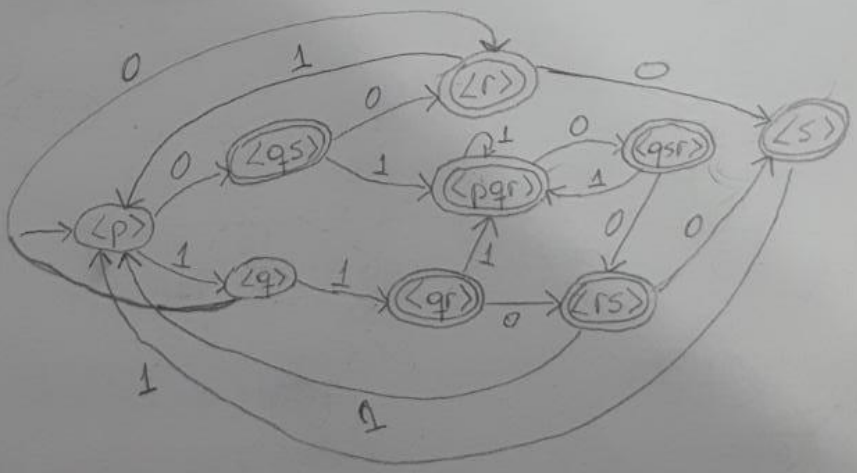
2.8)



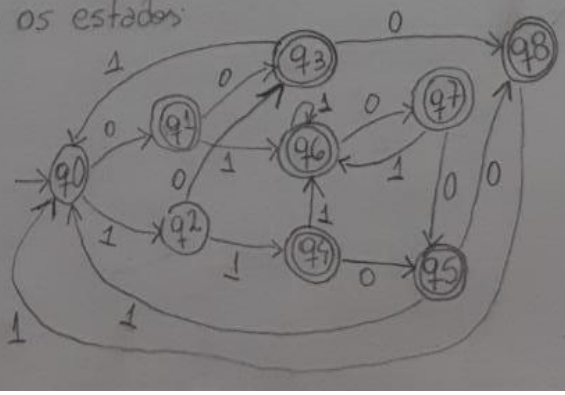
2.9)

δ	0	1
→ p	{q,s}	{q}
q	{r}	{q,r}
*r	{s}	{p}
*s	—	{p}

δ'	0	1
→⟨p⟩	⟨qs⟩	⟨q⟩
⟨q⟩	⟨r⟩	⟨qr⟩
*⟨qs⟩	⟨r⟩	⟨pqr⟩
*⟨r⟩	⟨s⟩	⟨p⟩
*⟨pqr⟩	⟨qsr⟩	⟨pqs⟩
*⟨qr⟩	⟨rs⟩	⟨pqr⟩
*⟨qsr⟩	⟨rs⟩	⟨pqr⟩
*⟨rs⟩	⟨s⟩	⟨p⟩
*⟨s⟩	—	⟨p⟩



renomeando os estados:



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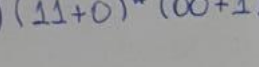
2.11 e 2.12

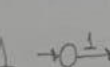
a) $(11+0)^*(00+1)^*$: w começa e/ou termina com 1's

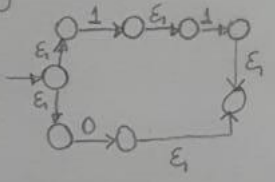
$(1+01+001)^*(\varepsilon+0+00)$: os 1's ou são seguidos de 0's ou seguido de ε.

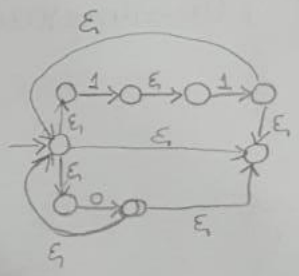
$(00+01+10+11)^*+((0+1)(0+1)(0+1))^*$: a linguagem aqui qualquer palavra desde que tenha pelo menos um 0

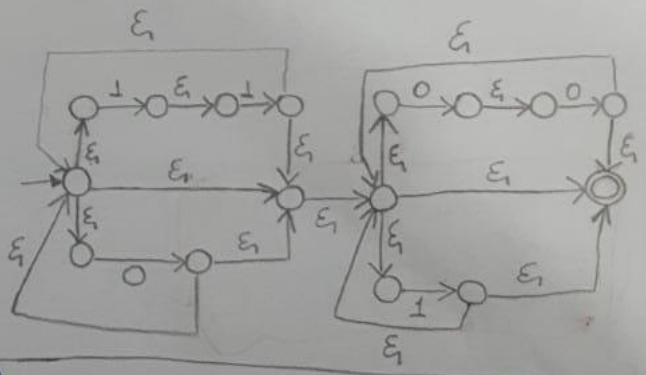
b) $(11+0)^*(00+1)^*$

11 

0 

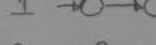
$(11+0)$ 

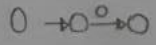
$(11+0)^*$ 

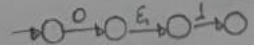


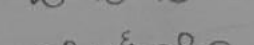
$(1+01+001)^*(\varepsilon+0+00)$

1 

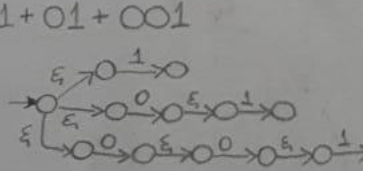
0 

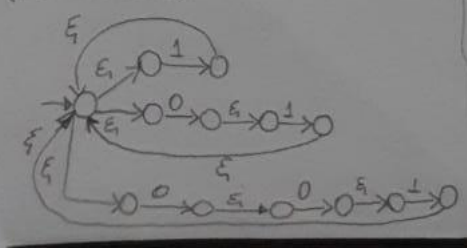
ε 

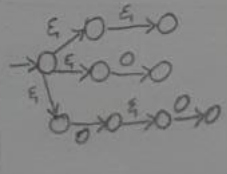
01 

00 

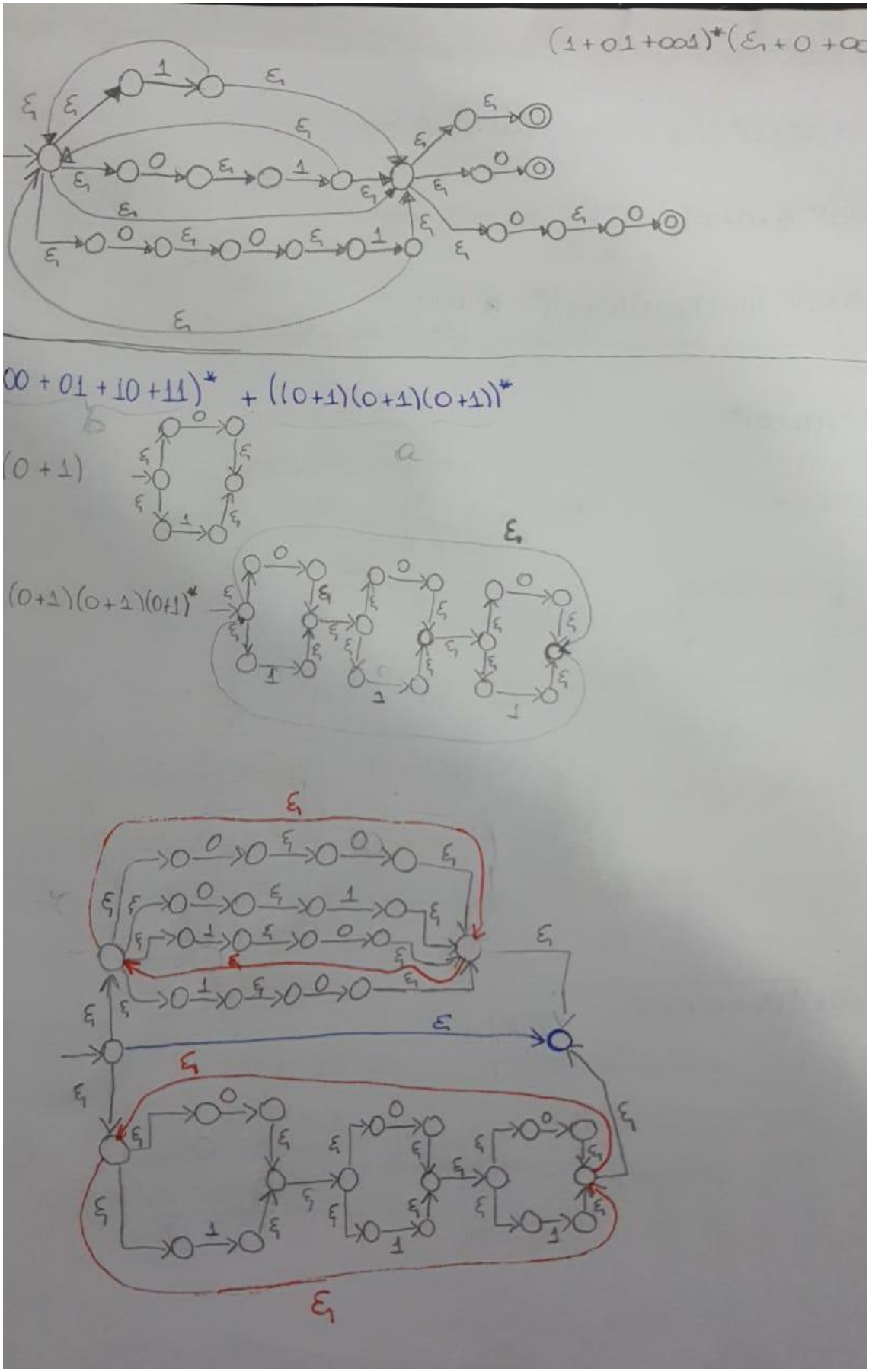
001 

$1+01+001$ 

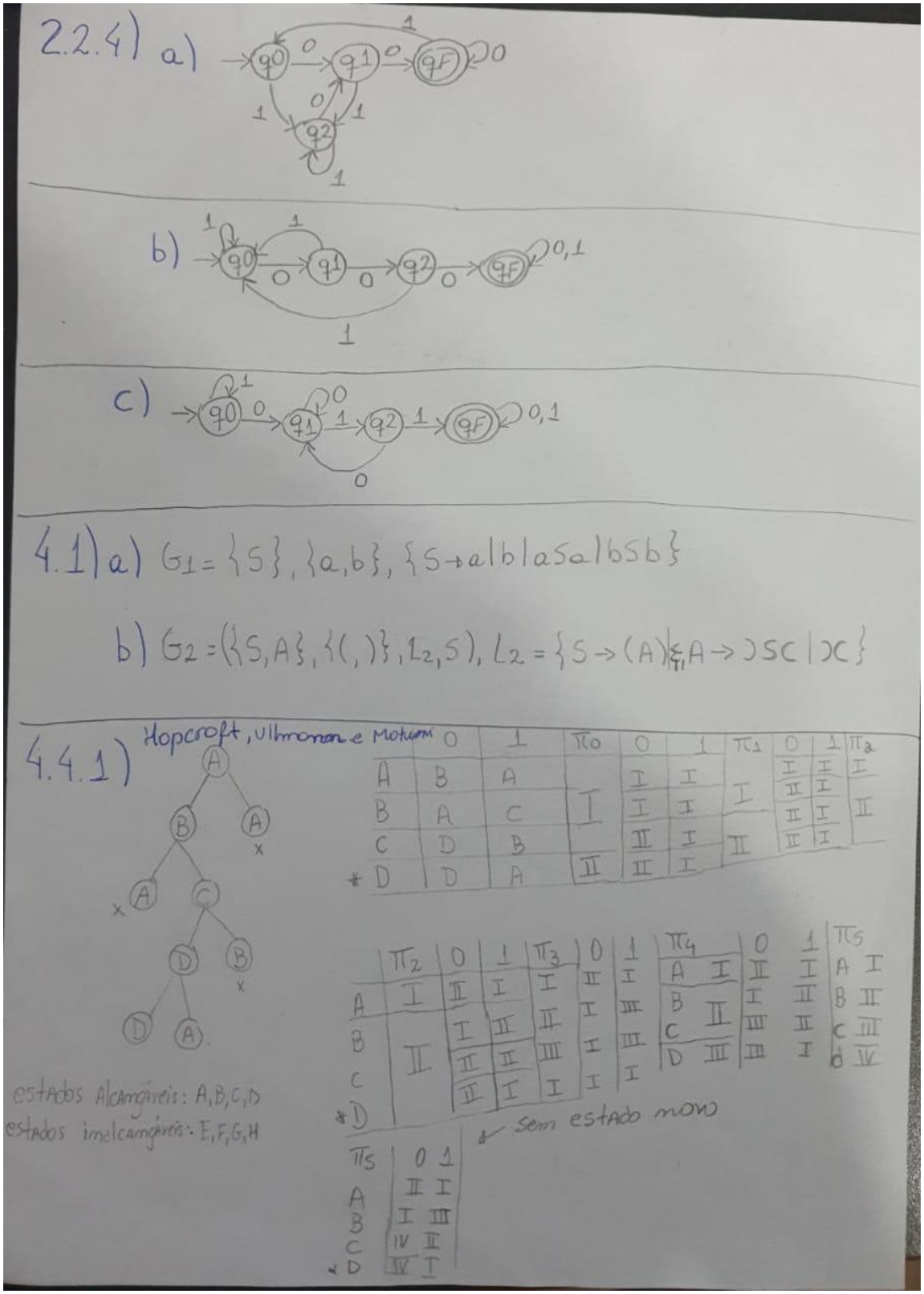
$(1+01+001)^*$ 

$\varepsilon+0+00$ 

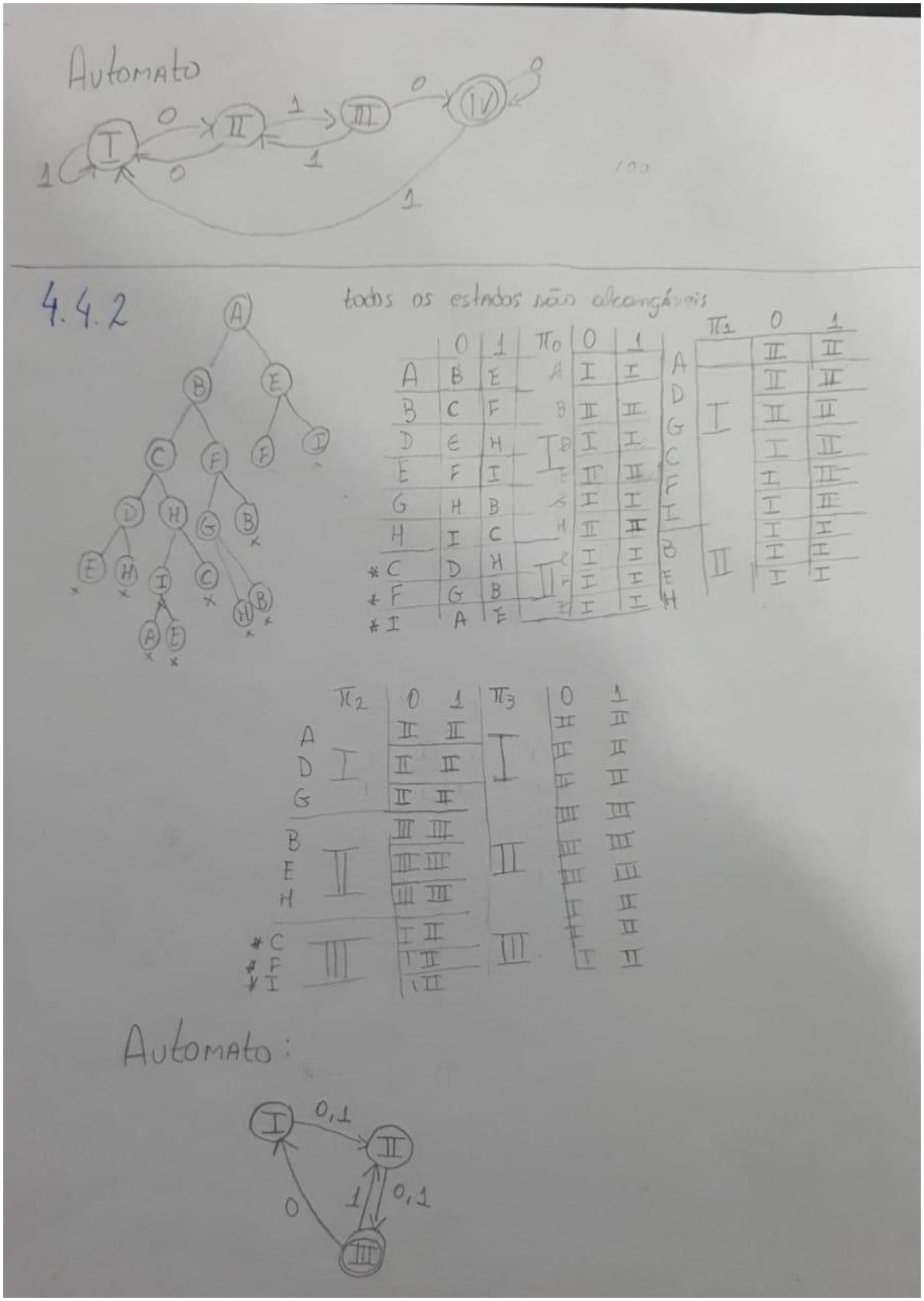
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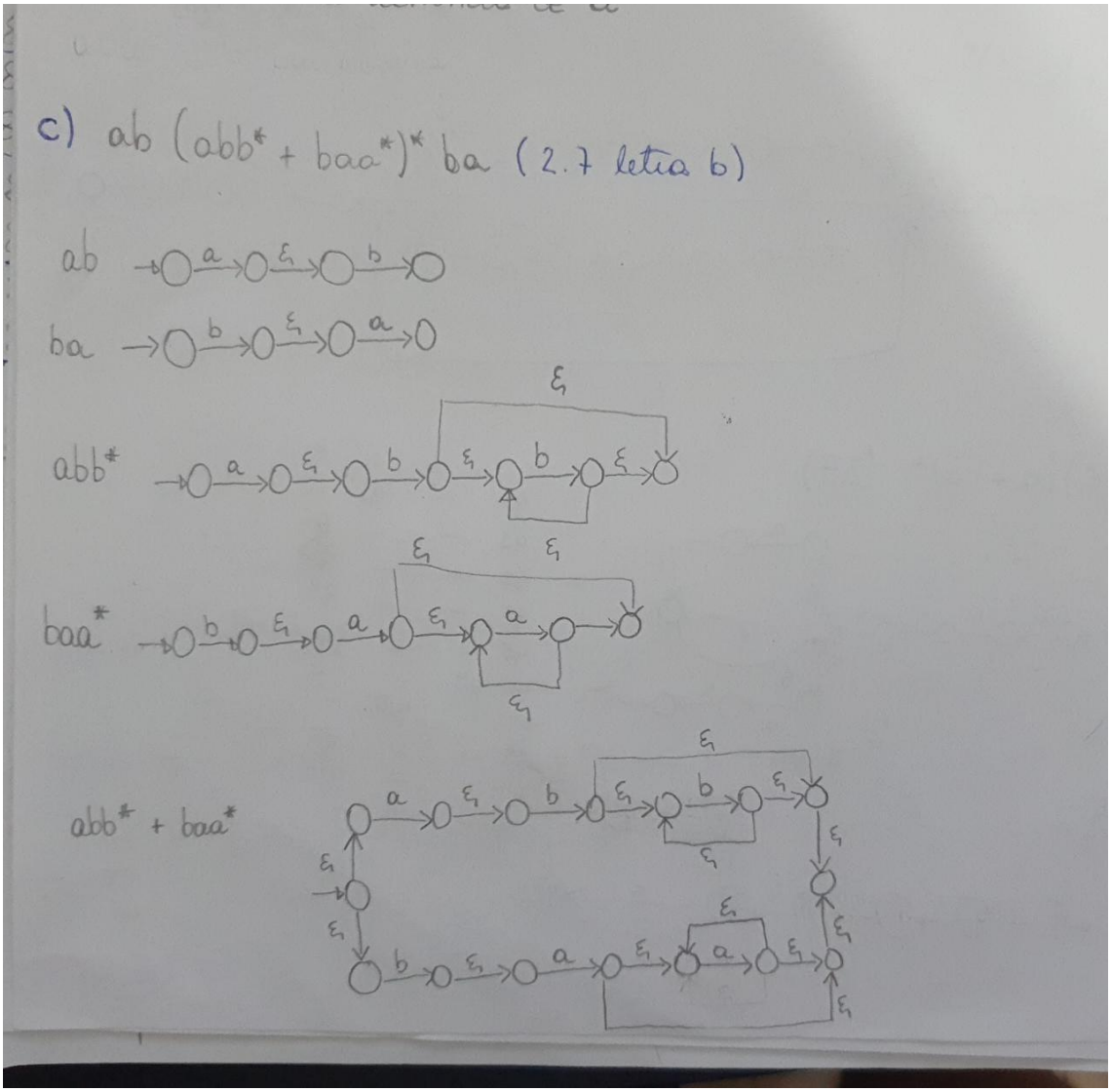
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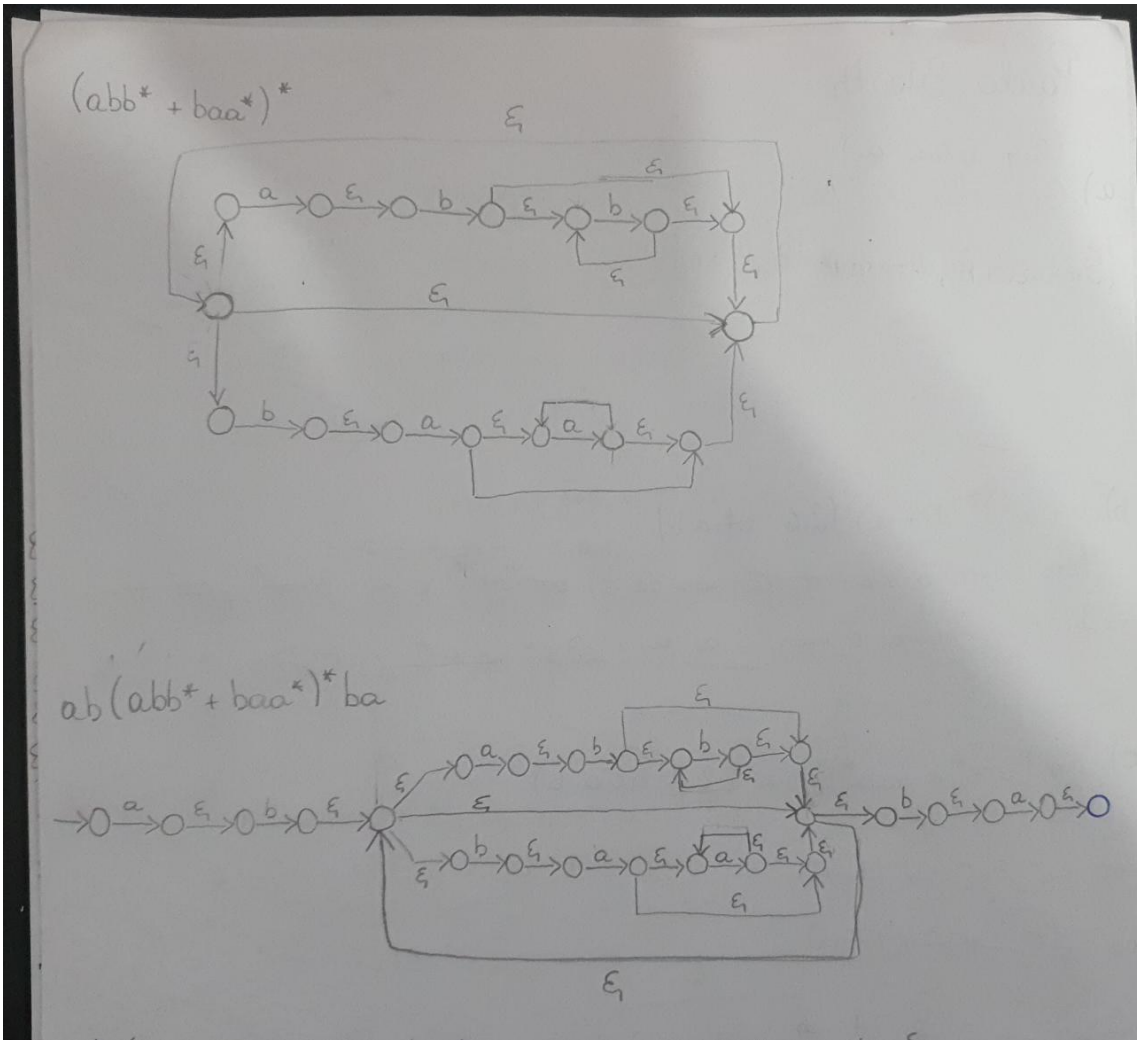
Livro Paulo Blauth:

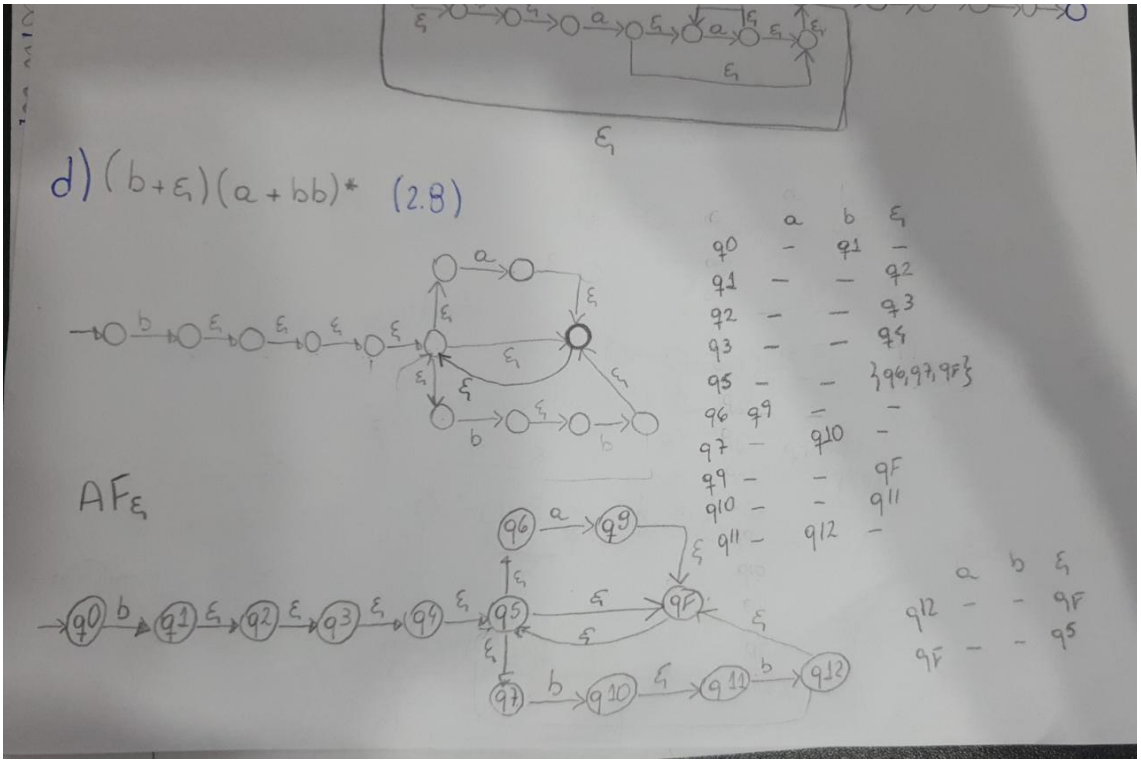
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a) (2.4) Optei por não fazer
b) (2.6) Optei por não fazer



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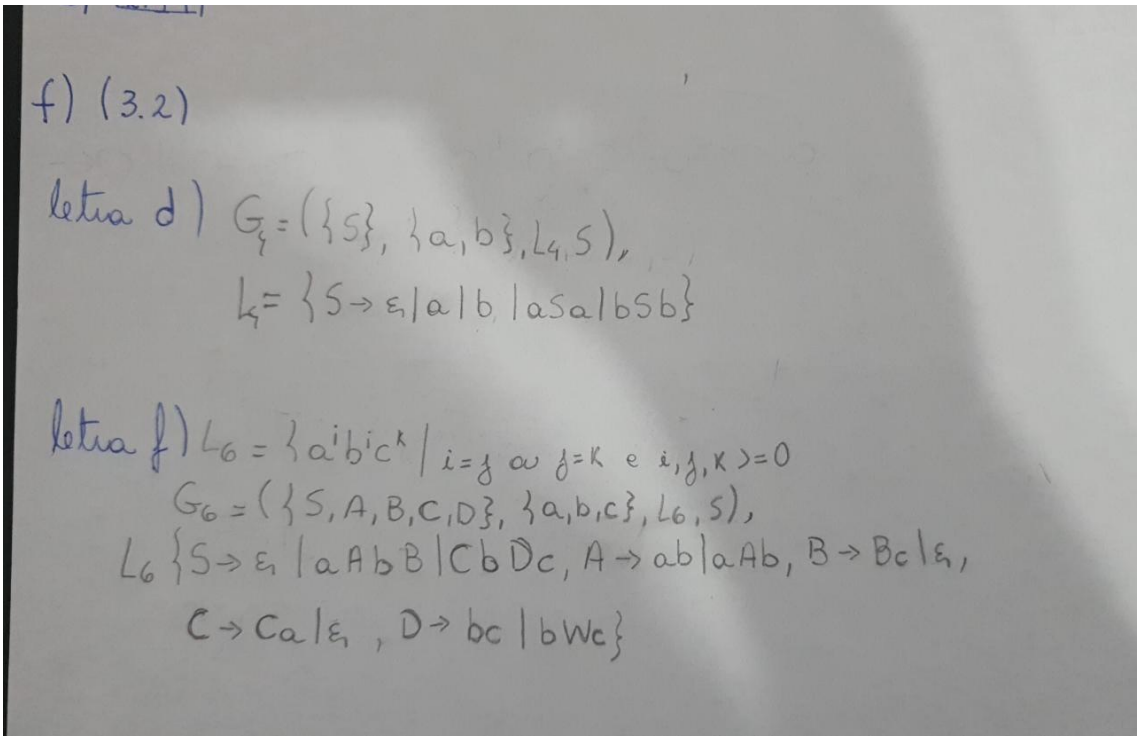


	a	b	ε
q0	–	q1	–
q1	–	–	q2
q2	–	–	q3
q3	–	–	q4
q5	–	–	{q6, q7, qF}
q6	q9	–	–
q7	–	q10	–
q9	–	–	qF
q10	–	–	q11
q11	–	q12	–

	a	b	ε
q12	–	–	qF
qF	–	–	q5

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e) (2.9) Optei por não fazer



g) (3.11) Optei por não fazer

h) (3.16) Optei por não fazer